

FOSSEE GUIVERSE HACKATHON 2026

**Visual Lab Pro:
Interactive Educational GUI &
STEM Simulation**

Platform: Scilab 2026.1.0

Problem Statement & Project Overview

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1. The Problem

In science, technology, engineering, and mathematics (STEM) education, there is a fundamental disconnect between abstract mathematical equations and physical reality. Students often struggle to internalize concepts like Ohm's Law, projectile kinematics, and exponential compound growth when they are presented solely as static formulas in textbooks or basic 2D plots. While advanced simulation software exists, it is often proprietary, resource-heavy, and not easily customizable.

Furthermore, within the Scilab ecosystem, the platform is primarily utilized for heavy numerical computation and static data visualization. There is a lack of open-source, highly interactive, modern graphical user interfaces (GUIs) that demonstrate Scilab's potential to act as a real-time, interactive educational engine.

2. Objective

The objective of **Visual Lab Pro** is to bridge the gap between mathematical abstraction and visual comprehension by transforming Scilab into a modern, interactive virtual laboratory.

Specific technical and educational objectives include:

- **Interactive Visualization:** To provide real-time, dynamic visual feedback for physics and electrical engineering principles. As users adjust environmental variables (e.g., voltage, launch angle), the mathematical consequences are rendered instantly.
- **Modern SPA Architecture in Scilab:** To engineer a Single Page Application (SPA) routing paradigm within Scilab, allowing seamless transitions between multiple laboratory modules without the memory overhead of spawning multiple windows.
- **Procedural Animation:** To bypass Scilab's single-threaded UI limitations by engineering thread-safe, finite-pulse simulation loops that animate concepts (like electron flow through a mathematically generated resistor) without freezing the application state.
- **Enterprise UI/UX:** To replace default environments with a custom-engineered, layered Dark Theme, providing a distraction-free, professional grade user experience.

3. Intended Users and Use Cases

- **STEM Students (High School & Undergraduate):** Can utilize the software to safely experiment with circuit loads, projectile parameters, and financial growth models to visually validate their textbook calculations.
- **Educators and Professors:** Can use the tool as a live classroom demonstration aid. The large UI elements, high-contrast dark theme, and real-time animations make it ideal for projector-based teaching.
- **Scilab Developer Community:** The open-source architecture serves as a high-level reference for developers looking to implement advanced GUI concepts in Scilab, including double-buffering, asynchronous UI reading, memory-safe routing, and procedural OpenGL geometry.

4. GUI Design & Application Architecture (Screenshots)

- **The Dashboard Interface** - The application acts as a central hub, utilizing a custom router to manage memory and navigate between isolated modules seamlessly.

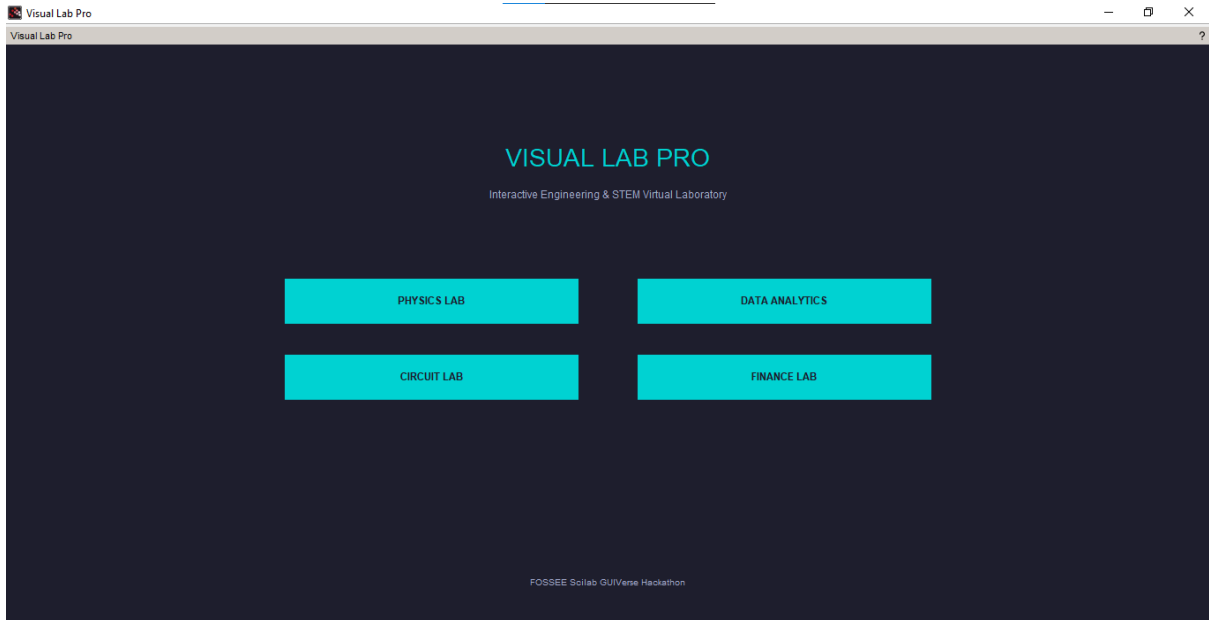


Figure 1: The main dashboard featuring the custom Java Swing dark theme and normalized scaling.

- **The Physics Laboratory** - This module visualizes projectile kinematics. It calculates predictive trajectories instantly as sliders are moved, rendering a dashed preview path before the simulation is fired.

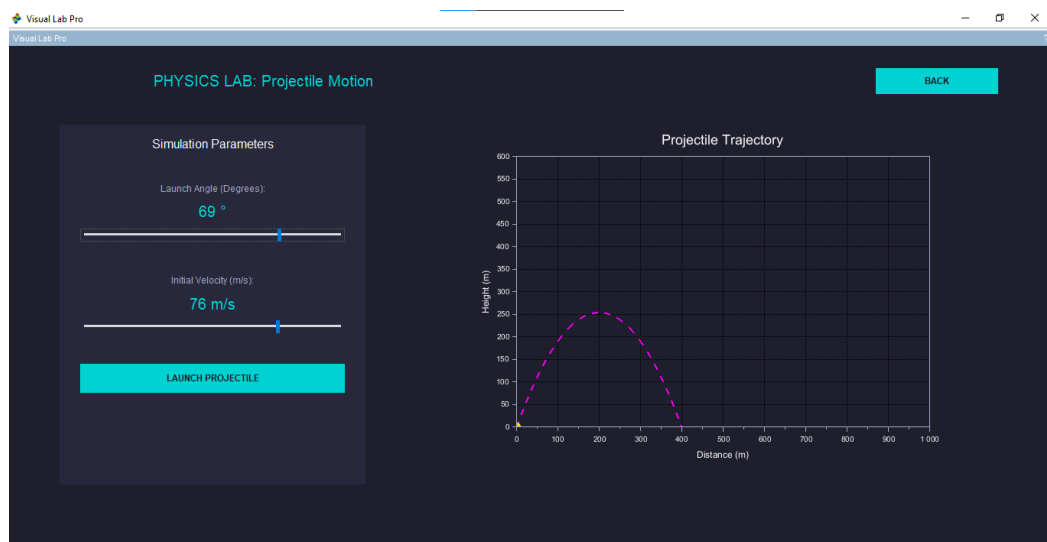


Figure 2: The Physics Lab interface displaying the interactive input panel and predictive dashed trajectory.

- **The Circuit Laboratory** - Instead of static images, this module features a procedurally drawn electrical schematic. In its idle state, it displays default parameter values, a perfectly rendered zig-zag resistor, and an open switch, ready for user interaction.



Figure 3: The Circuit Lab in its idle state, showcasing procedural OpenGL geometry and the custom control panel.

- **Analytics Data Laboratory** - Designed for pure statistical data visualization, this module demonstrates the engine's ability to render dynamic datasets, parse information, and execute UI layout updates smoothly.

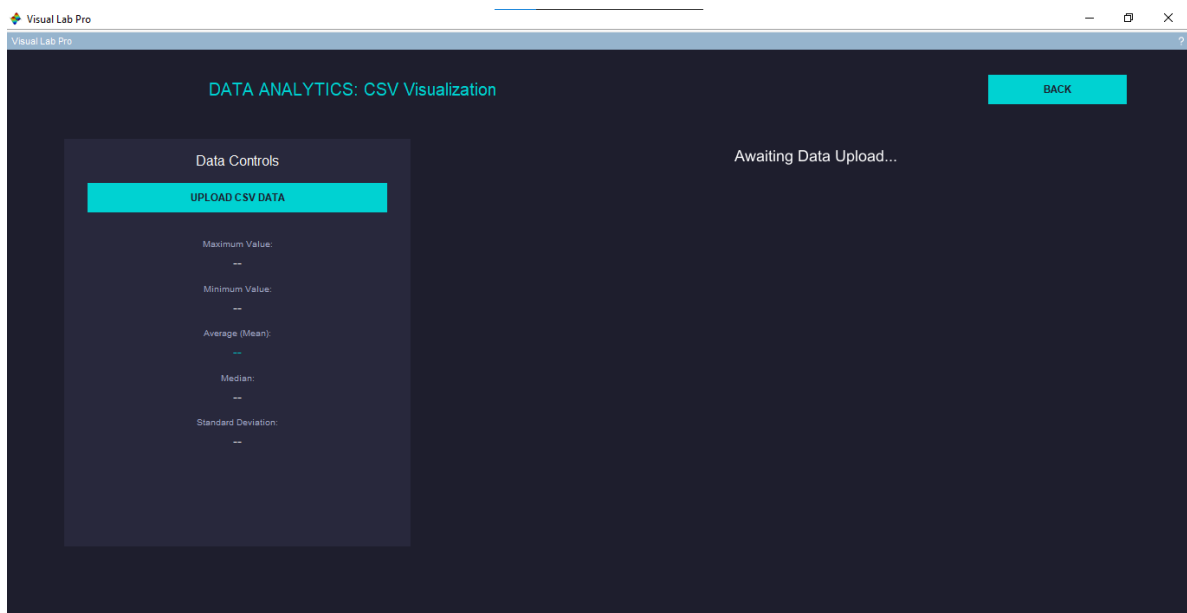


Figure 4: The Analytics Lab interface demonstrating data handling and dynamic axis scaling.

- **The Finance Laboratory** - Built for mathematical financial modeling, this interface allows users to tweak parameters like principal, interest rate, and time to visually comprehend complex compound growth without UI lag.



Figure 5: The Finance Lab interface displaying input sliders and the financial plotting canvas.

5. Sample Outputs & Interactive Results

- **Physics Lab:**

Real-Time Data Interception - We hijacked Scilab's default OpenGL tooltips to allow users to extract precise mathematical coordinates from generated geometries.

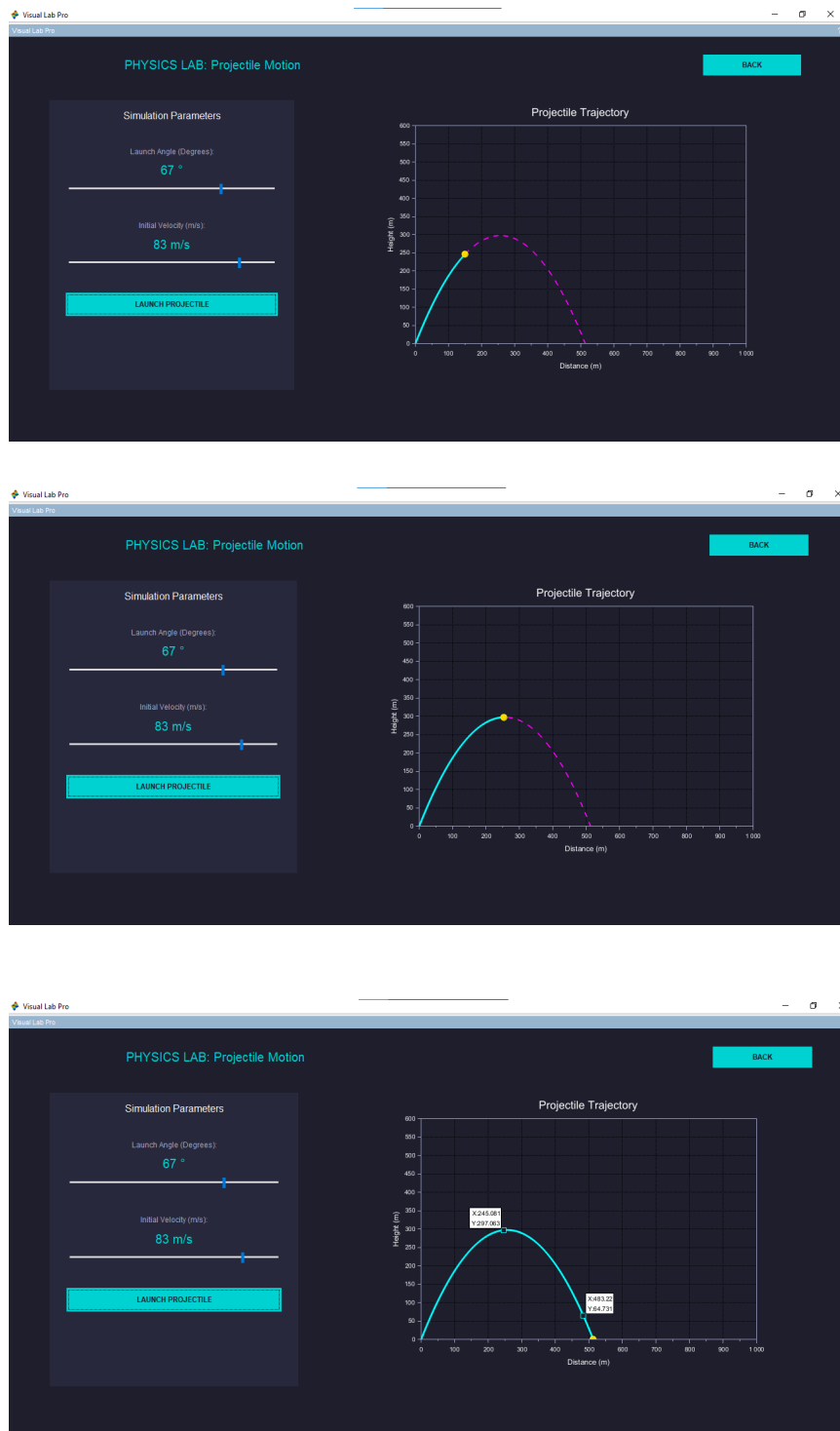


Figure 6: Sample output of a launched projectile, showing the projectile animation with interactive datatip successfully extracting the apex coordinates (X, Y max).

- **Circuit Lab:**

Thread-Safe Procedural Simulation - The Circuit Lab bypasses Scilab's single-threaded lockups by utilizing a finite-pulse engine. The schematic is drawn entirely with procedural math (including sine-wave resistor mapping).



Figure 7: The active circuit simulation. The software dynamically calculates power dissipation ($P = V * I$) and mathematically maps it to the RGB color matrix of the bulb, actively changing its glow while electrons trace the sine-wave resistor.

- **Analytics Lab:**

Live Statistical Output - By modifying the handle data directly in memory, the application can visualize dataset metrics and shape distributions instantly without computationally expensive re-draws.

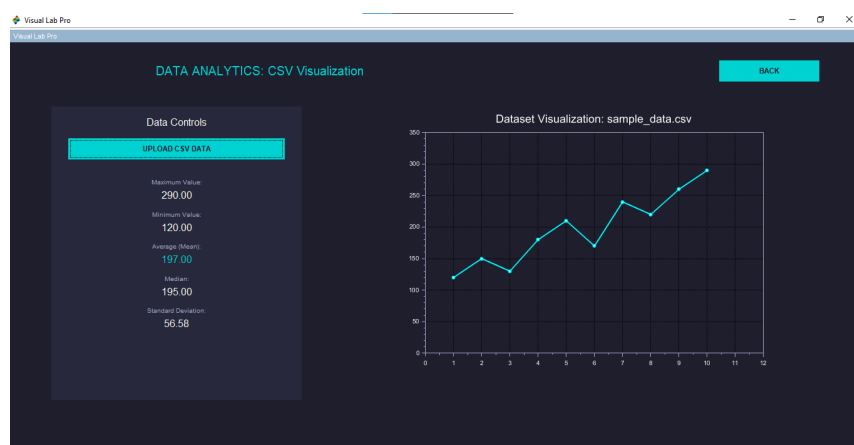


Figure 9: Sample output demonstrating real-time calculation and visualization of dataset distributions.

- **Finance Lab:**

Dynamic Growth Modeling - The finance engine instantly recalculates large-scale exponential formulas (removing messy scientific notations) and visualizes the compounding effect perfectly mapped to clean integer axes.



Figure 10: Sample output showing the resulting exponential growth curve and final maturity values calculated instantly upon slider mutation.